

BCA COLLEGE IDAR

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1. Exception Handling in Java

- **Exception:** An exception is an unwanted event that disrupts the normal flow of a program.
- **Exception Handling:** A mechanism to handle runtime errors so that normal execution of the program can continue

Types of Exceptions

1. Checked Exceptions

- Checked at **compile-time**.
- Must be handled using `try-catch` or declared using `throws`.
- Example: `IOException`, `SQLException`.

2. Unchecked Exceptions

- Occur at **runtime**.
- Not checked by compiler.
- Example: `ArithmeticException`, `NullPointerException`, `ArrayIndexOutOfBoundsException`.

Java Exception Handling Keywords

- **try** → block of code to test for exceptions.
 - **catch** → block to handle exception.
 - **finally** → block that always executes (cleanup code).
 - **throw** → used to throw an exception object explicitly.
 - **throws** → declares exceptions that a method can throw.
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Other Concepts

- **Nested try:** A `try` block inside another `try` block.
- **Multiple catch:** Multiple `catch` blocks for different exception types.
- **User-defined Exception:** Custom exception class created by extending `Exception` or `RuntimeException`.

Example:

Try

{

int a = 10/0;

}

catch(ArithmeticException e)

{

System.out.println("Divide by zero error!");

}

finally

{

System.out.println("This block always executes.");

}